INCH-POUND

MIL-S-46119C(MR)
26 October 1988
SUPERSEDING
MIL-S-46119B(MR)
4 April 1983

# MILITARY SPECIFICATION

## STEEL FORGINGS, TUBULAR PARTS FOR CANNON

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers alloy steel forgings that are produced by a hot plastic deformation process such as hammering, pressing, upsetting, extruding or any combination of these processes. This specification includes the minimum essential Engineering and Packaging Requirements and the necessary Quality Assurance Provisions to determine that these requirements have been met.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-C-13931 - Cannon, General Specification For

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, U.S. Army Laboratory Command, Materials Technology Laboratory, ATTN: SLCMT-MEE, Watertown, MA 02172-0001 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC NO. A3204 AREA FORG
DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:55Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

SWEWV/SARWV/SMCWV Form 230 - Report of Treatment and Test of

SARWV/SMCWV Form 1155 - Record of Treatment and Test of

(Note that the preface to these forms may be any one of the five letter organizational symbols listed above and separated by a "/".)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ASTM (AMERICAN SOCIETY FOR TESTING AND MATERIALS)

ASTM D3951 - Commercial Packaging, Practice for
ASTM E8 - Tension Testing of Metallic Materials
ASTM E23 - Notched Bar Impact Testing of Metallic Materials
ASTM E340 - Macroetching Metals and Alloys, Standard Method for
ASTM E381 - Macroetch Testing, Inspection, and Rating Steel Products, Comprising Bars, Billets, Blooms, and Forgings, Standard Method for

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103-1137.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 General requirements. Steel forgings shall be in accordance with the applicable drawings, the requirements of this specification, the applicable paragraphs of the General Cannon Specification - MIL-C-13931 as incorporated in this document, and all referenced documents. Any conflicts are to be resolved by means of the order of precedence (see 2.3).



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:55Z
Check the source to verify that this is the current version before use.

3.2 First Article. The contractor shall submit a first article unless it is specifically waived in the contract (see 4.3 and 6.2). No first article requirements shall be waived without review and approval by the procuring contracting officer (see 6.4).

3.2.1 First article waiver. The requirement for a first article may be waived on a new procurement of forgings of similar design and strength level to those previously procured from the same contractor. The waiver and determination of similarity will be made by the procuring contracting officer (see 6.2)

3.2.2 First article residual metal. Residual metal from the first article forgings shall be retained until the acceptance of the first article (see 6.2).

3.3 Chemical composition. Unless a particular composition of steel is specified on the drawing, in the invitation for bids, contract, or purchase order, (see 6.2) the contractor shall select a chemical composition which when heat treated shall meet the mechanical properties specified on the applicable drawing or contract. The contractor shall submit his proposed chemistry with his bid for forgings for informational purposes only. The contractor shall furnish a certified heat analysis for each forging.

3.4 Processing controls.

3.4.1 Orientation of forgings to ingot. Forgings shall be produced so that the longitudinal axis of the forging is parallel or coincident with the longitudinal axis of the ingot.

3.4.2 Straightening. Bows not exceeding 0.005 inch (arc to chord) per inch of length over which the bow extends may be removed by cold straightening after the final tempering treatment. If any bow exceeds 0.005 inch per inch, it shall be removed by hot straightening. The minimum temperature of the forging during the hot straightening operation is 600 degrees F (316 degrees C). The maximum temperature the forging can be heated to prior to the straightening operation is the final tempering temperature minus 50 degrees F (28 degrees C). Each forging that is hot straightened shall be retested for impact resistance properties after hot straightening.

3.4.3 Welding. Forgings shall not be welded except on prolongations. All welds shall be in such a position so as not to affect the finished tube or test metal.

3.5 Heat treatment. Heat treatment shall be performed on the whole forging (see 6.6.2), never on a portion only, and shall consist as a minimum of normalizing, austenitizing, agitated liquid quenching and tempering. Unless otherwise permitted by the procuring contracting officer (see 6.2), forgings shall be quenched vertically. Forgings for tubes with 57mm (2.24 inches) finished bore diameter and larger shall be rough bored and rough turned to the applicable forging drawing dimensions before heating and quenching. Unless otherwise allowed, the liquid coolant shall pass freely through the bore during the quenching process. Forgings for tubes with finished bore diameter under 57mm (2.24 inches) may be quenched solid at the option of the contractor. Tempering shall immediately follow quenching and



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:55Z
Check the source to verify that this is the current version before use.

shall consist of heating the entire forging to a temperature not less than 1000 degrees F (538 degrees C) for yield strength ranges up to and including 170,000 pounds per square inch (psi), nor less than 950 degrees F (510 degrees C) for yield strength in the range over 170,000 psi up to and including 190,000 psi, and holding at the temperature for a time appropriate to the thickness. A zone controlled tempering treatment employing different temperatures at opposite ends of the forging (differential tempering) may be used at the option of the contractor. Whenever differential tempering is employed, or whenever forgings are retempered only (see 4.6.3), temperatures of 1000 degrees F (538 degrees C) or 950 degrees F (510 degrees C) minimum are required, as noted, unless otherwise allowed by the procuring contracting officer, (see 6.2). The contractor shall submit his proposed heat treatment procedure with his bid for forgings for informational purposes only.

3.6 Mechanical properties.

3.6.1 Yield strength. Yield strength shall be as specified on the applicable drawings, or in the contract or purchase order. In addition, the spread in yield strength within any disc or between any two adjacent discs shall not exceed 6000 psi (see 4.5.3.1).

3.6.2 Reduction of area. The minimum percent reduction of area for individual specimens shall be as specified in table I (see 4.5.3.2).

3.6.3 Charpy V-notch impact resistance. The minimum value for individual Charpy V-notch impact resistance specimens shall be as specified in table I (see 4.5.3.3), for the average yield strength obtained (see 6.6.1).

TABLE I. Mechanical properties.

| Yield strength (0.10 percent offset) psi | Minimum reduction of area; single value percent | | Minimum Charpy V-notch impact resistance at -40 ± 2°F (-40 ± 1°C); single value, foot pounds | |
|---|---|---|---|---|
| | Trans. | Long | Trans. | Long. |
| 120,001 - 130,000 | 40 | 50 | 35 | 45 |
| 130,001 - 140,000 | 35 | 45 | 30 | 40 |
| 140,001 - 150,000 | 30 | 40 | 25 | 35 |
| 150,001 - 160,000 | 30 | 40 | 20 | 30 |
| 160,001 - 170,000 | 25 | 35 | 15 | 25 |
| 170,001 - 180,000 | 25 | 35 | 15 | 20 |
| 180,001 - 190,000 | (See applicable forging requirements) | | | |

NOTE: See paragraph 6.8 for metric conversion table.

3.7 Macrostructure. Macrostructure of forgings shall be equivalent to or better than Standards S-2, R-2, and C-3 of ASTM E381, Plate I and shall also meet the requirements of Plate II. Standard C-3 only applies when the forging is heat treated solid (see 3.5).



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:55Z
Check the source to verify that this is the current version before use.

3.8 Dimensions and surface roughness. Dimensions and surface roughness, and dimensional tolerances shall be as specified on the applicable forging drawing.

3.9 Identification marking. Each forging shall be marked on the breech end, by metal-die stamping, with the heat, forging serial number, drawing number, and the manufacturer's identification. When special identification marking is required, see 6.2.

3.10 Records. The contractor shall record the information needed to comply with this specification to enable the Government representative to verify and sign SWEWV/SARWV/SMCWV Form 230 or SARWV/SMCWV Form 1155.

3.11 Workmanship. The forging shall be free from defects; such as laps, seams, cracks, etc.; which cannot be removed by machining the forging to its finished dimensions. The existence of these defects shall cause the forging to be rejected even if all other requirements of this specification are met.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. first article inspection (see 4.3).

b. Quality conformance inspection (see 4.5).

4.3 First article inspection.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:55Z
Check the source to verify that this is the current version before use.

4.3.1 General. Unless otherwise specified in the contract (see 6.2), a first article sample shall be submitted for inspection. The first article sample shall be representative of the production processes to be used during quantity production.

4.3.2 First article sample. Unless otherwise specified in the contract, the first article sample shall consist of one forging (see 6.2).

4.3.2.1 Inspection provisions. The first article sample shall be subjected to quality conformance inspection (see 4.5), additional inspection specified herein, and the provisions of the contract. A minimum of one (1) forging in the sample shall be destructively tested by taking specimens at each end and at 1/4, 1/2 and 3/4 of the length of the forging unless otherwise specified. The number of specimens and tests required for each of these locations shall be as shown in table II.

4.4 Number of specimens and tests. Unless otherwise specified on the applicable drawing or in the contract or purchase order, (see 6.2) the number of specimens and tests for individual or multi-length forgings shall be as shown in table II.

TABLE II. Number of specimens required for each test.

| Test | Number of Specimens | |
|---|---|---|
| | Individual forgings | Multi-length forgings |
| Macrostructure | 1 - each end | 1 - at each end and<br>1 - at each junction between adjoining lengths |
| Tension | 2 - each end | 2 - each end and<br>2 - at each junction between adjoining lengths |
| Charpy V-notch | 2 - each end | 2 - at each end and<br>2 - at each junction between adjoining lengths |

4.5 Quality conformance inspection. All forgings shall be subjected to the quality conformance inspections as specified in table III which lists inspections and methods and their corresponding requirements.

TABLE III. Quality conformance inspection.

| Examination or test | Requirement paragraph | Examination or test paragraph |
|---|---|---|
| Yield strength | 3.6.1 | 4.5.1.3.1<br>4.5.1.4.1<br>4.5.3.1 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:55Z
Check the source to verify that this is the current version before use.

TABLE III. Quality conformance inspection. (Continued)

| | | |
|---|---|---|
| Reduction of area | 3.6.2 (table I) | 4.5.3.2 |
| Charpy V-notch | 3.6.3 (table I) | 4.5.1.3.2<br>4.5.1.4.2<br>4.5.3.3 |
| **Examination or test** | **Requirement paragraph** | **Examination or test paragraph** |
| Macrostructure | 3.7 | 4.5.2.1<br>4.5.2.2 |
| Identification marking | 3.9 | 4.10 |
| Workmanship | 3.11 | 4.11 |
| Packaging | 5. | 4.12 |

4.5.1 Mechanical properties.

4.5.1.1 Location of discs. All discs from quenched and tempered forgings shall be located at a distance of at least one wall thickness from the end of the forging (see 6.6.2). After discs have been taken, in accordance with the above requirements, additional thermal treatment of the forging shall be limited to retempering, hot straightening, or retempering and hot straightening (see 3.4.2). Additional discs from each end are required adjacent to the previous discs to each provide two tension and two Charpy impact specimens after any forging has been reheat treated or hot straightened. Test values shall comply with the applicable forging drawing and table I.

4.5.1.2 Location of test specimens.

4.5.1.2.1 Transverse. Tension and Charpy V-notch impact test specimens shall be taken from the location indicated on the applicable drawing. Specimens shall be taken so that the longitudinal center lines of the specimens are in a plane approximately perpendicular to the longitudinal axis of the forging. Charpy V-notch specimens shall be notched on the side nearest the bore surface. Breech end and o'clock positions shall be maintained for each disc. Tension specimens shall be taken at the 12 and 6 o'clock positions. Charpy impact specimens shall be taken at the 3 and 9 o'clock positions. If additional specimens are required, they shall be taken from the same disc or an adjacent disc when sufficient material is not available.

4.5.1.2.2 Longitudinal specimens. When the forging design precludes the use of transverse specimens, longitudinal specimens shall be used, in which case the specimens shall be taken with their longitudinal centerlines approximately parallel to the longitudinal axis of the forging. Charpy specimens shall be notched on the side nearest the bore surface. The positions of the tension and Charpy specimens shall be the same as in 4.5.1.2.1.



Source: https://assist.dla.mil -- Downloaded: 202640-22T04:55Z
Check the source to verify that this is the currentversion before use.

4.5.1.3 Test specimens.

4.5.1.3.1 Tension. Tension test specimens shall be machined to the form and dimensions of the standard 0.250, 0.350, 0.500 inch diameter test specimen with 1.000, 1.400, or 2.000 inch gage lengths respectively, as shown in ASTM E8.

4.5.1.3.2 Charpy V-notch impact. Charpy V-notch impact specimens shall be machined to the form and dimensions of the standard 0.394 inch by 0.394 inch by 2.165 inch specimen shown in ASTM E23.

4.5.1.4 Test methods.

4.5.1.4.1 Tension. Yield strength shall be determined by the offset method in accordance with ASTM E8. The limiting offset shall be 0.10 percent (0.001 inch per inch of gage length.) The rate of stress application when determining the yield strength shall not exceed 100,000 pounds per square inch per minute.

4.5.1.4.2 Charpy V-notch impact. Charpy V-notch impact tests shall be conducted in accordance with ASTM E23. The temperature of the test specimens shall be minus 40 ± 2 degrees F or -40 ± 1 degree C.

4.5.2 Macrostructure.

4.5.2.1 Test discs. Discs shall be cut from each end of the forging for macroscopic inspection. The discs shall be taken from metal at least one wall thickness from the end of the forging as originally heat treated. At the option of the contractor, the discs may be those taken from each end for mechanical testing. Both discs must meet the requirements of 3.7.

4.5.2.2 Test method. Discs shall be prepared and examined in accordance with ASTM E340 and ASTM E381 to determine compliance with 3.7.

4.5.3 Interpretation of test results.

4.5.3.1 Yield strength. The forging shall have met requirements for yield strength if all values of yield strength fall within the limits prescribed by the applicable drawings and the requirements of 3.6.1.

4.5.3.2 Reduction of area. The forging shall have met the requirements for reduction of area if all values of the reduction of area are equal to or greater than that shown in table I. The minimum percent reduction of area is shown opposite the applicable yield strength range (see 3.6.2).

4.5.3.3 Charpy V-notch impact resistance. The forging shall have met the requirements of Charpy V-notch impact resistance if all values of the impact resistance are equal to or greater than that shown in table I. The minimum foot-pounds resistance is shown opposite the applicable yield strength range, into which the average yield strength falls (see 3.6.3 and 6.6.1).

4.6 Rejection. Forgings which fail to meet the requirements of this specification when tested in accordance with 4.5 shall be rejected.



Source: https://assist.dla.mil -- Downloaded: 202640-22T04:55Z
Check the source to verify that this is the currentversion before use.

4.6.1 Local defects. If rejection of a forging can be traced to the presence of allowable defects (see 3.7) in the failed test specimen, the test values obtained on that specimen may be disregarded subject to the approval of the procuring contracting officer. Two wake specimens shall be substituted for the defective specimen. Only one such substitution is permitted per forging. When the wake specimens have been substituted, failure of the forging to meet specification requirements based on test results from the wake specimens shall be cause for rejection of the forging, whether or not the wake specimens also exhibit defects.

4.6.2 Defective machined specimens and faulty testing equipment. If defectively machined test specimens or faulty operation of the testing machine or recording device causes the rejection of a forging, the test values obtained on such specimens may be disregarded subject to the approval of the procuring contracting officer, and another specimen from the same forging end may be substituted for each such specimen.

4.6.3 Treatment for retest. Forgings that do not meet the yield strength, reduction of area or Charpy V-notch resistance requirements of this specification except as noted in 4.6.1 and 4.6.2, shall be reheat-treated before retesting. Reheat treatment may consist of retempering or reaustenitizing, agitated liquid quenching, and retempering. All retempering without reaustenitizing shall be at a temperature equal to or greater than the original tempering temperature for the forging.

4.7 Check tests. From any lot of forgings found acceptable under prescribed inspection, the Government may designate any forging as a check forging. Such a forging shall be tested at a facility to be designated by the Government, and shall be subjected to any or all tests specified herein. If the forging fails to comply with the requirements of the applicable engineering drawing and table I, two additional forgings shall be tested. If both forgings comply with all requirements, the lot shall be accepted. If either forging fails to comply with all the requirements, the lot shall be rejected.

4.7.1 Check test residual metal. Residual metal from any check forgings shall be retained until the check forgings are either accepted or rejected (see 6.2).

4.8 Records. The results of all tests taken on each forging shall be recorded. These records shall include results of tests taken before and after retreatment and the discarded results of any tests with the reasons given for the discard. All heat treatment cycles and results of all tests taken on each forging shall be reported on SWEWV/SARWV/SMCWV Form 230 or SARWV/SMCWV Form 1155.

4.9 Final acceptance. Final acceptance of forgings shall be made when the final inspection satisfactorily establishes their compliance with the requirements of this specification.

4.10 Identification marking. Each forging shall be visually examined for identification marking. Failure to comply with 3.9 shall be cause for rejection.



Source: https://assist.dla.mil -- Downloaded: 202640-22T04:55Z
Check the source to verify that this is the currentversion before use.

4.11 Workmanship. Each forging shall be visually examined for workmanship. Failure to comply with 3.11 shall be cause for rejection.

4.12 Packaging. Packaging, packing and marking for shipment shall be visually examined. Failure to comply with the requirements of Section 5 shall be cause for rejection.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified, the forgings shall be packed for shipment in accordance with ASTM D3951.

5.2 Marking for shipment. Marking for shipment shall be in accordance with ASTM D3951 in addition to any special marking required in the contract or purchase order.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The steel forgings specified herein are intended to be used as tubular parts of military cannon such as tubes, jackets, and liners.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Instructions for the submission of a first article, including the number of forgings, unless the first article is waived (see 3.2, 4.3 and 6.4).

d. Residual metal (see 3.2 and 4.7).

e. Steel composition if required (see 3.3).

f. Heat treatment to be used if required (see 3.5).

g. Applicable drawings or sketches showing locations and numbers of test specimens, and mechanical property requirements (see 3.6 and 4.4).

h. Special identification marking not covered in 3.9.

i. The examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

j. All inspection and test data that is required including any additional or extended examinations and tests beyond the scope of this specification.



Source: https://assist.dla.mil -- Downloaded: 20260-22T04:55Z
Check the source to verify that this is the current version before use.

k. At whose expense a retest may be performed.

l. Inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified.

m. Availability of inspection equipment from the Government.

n. Extent of contractor's responsibility for Government furnished and for contractor-required final inspection equipment.

o. Applicable acceptance test procedures.

p. Packing and marking for shipment (see 5.1 and 5.2).

q. When warranted, the application of MIL-I-45208.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

| Reference Paragraph | DID Number | DID Title | Suggested Tailoring |
|---|---|---|---|
| 4.3 | DI-T-4902 | First Article Inspection Report | --- |
| 3.3 | DDI-T-23264 | Certification Data Report | --- |

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.3.1 Reports. Copies of SWEWV/SARWV/SMCWV Form 230 and SARWV/SHCWV Form 1155 should be obtained from the procuring contracting officer.

6.4 First article. When a first article inspection is required, the procuring contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first (how many) production items, a standard production item from the contractor's inventory, and the number of items to be tested as specified in 4.3. The procuring contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders



Source: https://assist.dla.mil -- Downloaded: 202640-22T04:55Z
Check the source to verify that this is the currentversion before use.

offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Waivers and deviations. The procuring contracting officer shall coordinate all requests for waiver or deviation to this specification with Benet Laboratories and the Product Assurance Directorate at Watervliet Arsenal.

6.6 Definitions.

6.6.1 Average yield strength. The average yield strength of the forging under consideration is the arithmetical mean of all yield strength values obtained from the same end at the stage of heat treatment at which tests are being performed.

6.6.2 Forging. A forging for purposes of inspection and test is defined as the forging, the length of which is shown on the applicable rough-machined forging drawing, plus any prolongations as defined in 6.6.3 which are integral parts of the piece as heat treated. In the case of forgings formed and heat treated in multiple lengths, one set of specimens (in addition to the set of specimens obtained from each end of the multiple forging) shall be obtained from the test metal between each pair of adjoining forgings to be applicable to both forgings. Provisions shall be made for identification of test specimens with the particular forging within a multiple forging.

6.6.3 Prolongation. A prolongation for purposes of inspection and test is defined as an extension beyond the rough-machined forging length. The diameters and mass distribution of the prolongation shall not be less than those of the forging at a point coincident with the end of the rough-machined forging, from which it extends.

6.6.4 Retest. A retest is defined as any test made after the results of the original test are known.

6.6.5 Test metal. Same as Prolongation (see 6.6.3).

6.6.6 Wake bar. A wake bar is a test specimen which is substituted for another specimen which has been discarded because of the presence of local defects. The wake bar should be taken as close as possible to the original specimen.

6.7 Finish-machined contour. The contour of the finish-machined tube should be indicated on the forging drawing or on a separate drawing furnished by the procuring contracting officer.

6.8 Metric units. When metric units are required, units for degree Fahrenheit, pounds per square inch, square inch, inch, and foot-pound may be converted to the metric equivalent by multiplying them by the following conversion factors:



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:55Z
Check the source to verify that this is the current version before use.

| English | Multiply by | Equals | Metric SI Unit |
|---|---|---|---|
| degree Fahrenheit | (F-32)x5/9 | = | degree Celsius (C) |
| pounds per square inch | 6.895E3 | = | Pascal (Pa) |
| square inch | 6.452 | = | square centimeter |
| inch | 2.54 | = | Centimeter (cm) |
| foot-pound | 1.356 | = | Joule (J) |

Note: Conversion factors can be associated with ASTM E380 entitled "Metric Practice Guide".

6.9 Drawings. Drawings listed in Section 2 of this specification under the heading "U.S. Army Armament Research, Development and Engineering Center" (ARDEC) may also include drawings prepared by, and identified as Watervliet Arsenal, Ordnance Corps, Weapons Command, or U.S. Army Armament Research and Development Command, U.S. Army Armament, Munitions and Chemical Command, etc. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.10 Subject term (key word) listing.

Cannon
Forgings
Tubular parts
Steel

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.12 International standardization agreement. Certain provisions of this specification 4.5.1.3 and 4.5.1.4 are the subject of international standardization agreements QUADRIPARTITE STANDARDIZATION AGREEMENT - 129 (Edition 2) and NATO-STANAG 4020. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

6.13 Beneficial comments. Duplicate copies of beneficial comments addressed to Director, U.S. Army Materials Technology Laboratory, ATTN: SLCMT-MEE, Watertown, MA 02172-0001 should also be sent to Director, Benet Laboratories, ATTN: SMCAR-CCB-SAS, Watervliet, NY 12189-4050.

Custodian:
Army - MR

Review activity:
Army - AR

User activity:
Army - ME

Preparing activity:
Army - MR

Project FORG-0158

(KBWP# ID-0246A/DISK 0138A. FOR MTL USE ONLY)



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:55Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-S-46119C(MR) | STEEL FORGINGS, TUBULAR PARTS FOR CANNON |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:55Z
Check the source to verify that this is the current version before use.

...standardization documents better, the DoD provides this form for use in